

Template for FEDR Report

Use the following as a guideline when drafting the FEDR. You may add, remove, or rearrange content to create a clearer, more compelling technical narrative. However, please keep all Chapter and section titles as these are needed for consistent assessment. Any further sub-sections are entirely up to your design team.

The inclusion of Figures and Tables in the report is highly encouraged. Minimum font size of text should be 11 pt Aptos with 1.15 line spacing and at least 1" margin on all 4 sides of an A4 page. The page counts indicated are the absolute maximum; please stay well within that.

Chapter	Section	Description	Page Count	Remarks / Comments
---	---	Cover Page	1 page max	Include group no, group name (if any) & group members
---	---	Executive Summary	1 page max	Final concept, key performance figures, and why it was chosen.
---	---	Table of Contents (ToC)	As needed	Plus table of abbreviations/symbols if desired
1	Introduction		4 pages max (total for Ch-1)	---
	1.1	Project Context & Mission Requirements		Summary of design brief, key mission requirements, customer needs and constraints
	1.2	Overview of the Final Concept		High-level description of chosen configuration, key features and expected performance against customer requirements
	1.3	Structure of report		Short roadmap explaining how the report is organised
2	Initial Sizing		6 pages max (total for Ch-2)	
	2.1	Concept down-selection methodology		Process used to select your final two concepts for comparison. MCDA tables can be included in an appendix. This section should instead describe the process, engineering rationale, and outcome.
	2.2	High-level description of the sizing tool		Give a high-level description of the sizing tool used to quantify the performance of your final two concepts. The aim is not to be comprehensive here. State hyper-parameters, why they've been selected, and give an impression of the key design drivers required for inclusion into the sizing tools to allow for a fair comparison between the two concepts. (e.g., wing mass and aerodynamic drag estimates for a truss-braced wing) The aim here is to help the reader understand your rationale for your final concept decision. Detailed description of methodology comes in the next chapter
	2.3	Design Space exploration		Overview of the trade studies used to select for the final concept.

Chapter	Section	Description	Page Count	Remarks / Comments
3	Detailed Sizing Methodology		Approx. 4 pages per-section (max of 32 pages total for Ch-3)	--- This section describes, in detail, the sizing tool developed for your final concept.
	3.1	Overview		Overview of the final methodology, including key hyperparameters, discipline dependencies, and overall architecture.
	3.2	Mission and Constraint Analysis		Each section should detail the methodology developed within the associated discipline and provide discipline-specific validation/calibration used to tune the numerical tool.
	3.3	Wing Aerodynamics		
	3.4	Stability and Control		For guidance on the content of each section, please refer to the Unit Handbook. The fidelity of final methods may vary between groups and will depend on the difficulty of the chosen concept. We are looking for the use of an appropriate, physics-informed fidelity that captures the key design drivers of your chosen concept.
	3.5	Fuselage Structures*		
	3.6	Wing Structures		
	3.7	Landing Gear*		
	3.8	Direct Operating Cost Model		
	3.9	Climate Impact Model		*Note: If there are no fuselage structures or landing gear engineers in your group, state this in the relevant section and then state the simplified model used instead.
4	Design Optimisation, Trade Studies, and Sensitivity Studies		13 pages max (total for Ch-4)	
	4.1	Model Validation		Outline comparisons made to assess the validity of the developed sizing tool (e.g. validation against baseline aircraft)
	4.2	Exploration		Provide a detailed exploration of the design space. For example, show global Pareto fronts, conduct trade-off/sensitivity studies, and examine different points in the design space to highlight critical hyperparameters, interdisciplinary dependencies, and interesting sensitivities.
	4.3	Selection of the final design		Rationale for the selection of a final design in the face of multiple design objectives. Comment on the robustness of the chosen design and the influence of competing disciplines on the chosen design.
	4.4	Future Development		State the largest assumptions and uncertainties in your current design tool. Use FHA to rank their risk to the future development of your concept and provide guidance for a future

Chapter	Section	Description	Page Count	Remarks / Comments
5	Project Management		5 pages max (total for Ch-5)	---
	5.1	Project Management Philosophy		Detail the process and tools used to manage the team during the project, including task management and maintenance of shared files.
	5.2	Continued Development		Discuss how management techniques evolved over the course of the project and reflect on the key strengths and weaknesses of the final strategy and how you would change it for a future sprint.
6	Summary / Conclusions		1 page max	---
Total Size (not including references and appendices)			66 pages max	
---	---	References	As needed	Index in order of citations in the report
---	---	Appendices - (A,B,C...)	Unlimited	As applicable, may be referred but will not be assessed